

A Productivity Fixated World in the Age of AI: Investigating The Purpose of Human Art

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Abstract

Modern society increasingly prioritizes final products over the creative process, a trend exacerbated by the rapid adoption of generative AI. This study aims to explore the impact of AI integration into the creative process on subjective well-being and fulfillment, and to determine whether these effects differ between individuals creating for leisure versus those creating for a profession. As a part of this research, a quantitative-focused survey was conducted with 200 working adults, categorized using a 2x2 design: acting as either a hobbyist or a professional illustrator, and either utilizing AI or not utilizing AI. The data reveals a consistent and statistically significant drop in a person's well-being, across both hobbyist and professional groups when AI is utilized in the creative process. The differences in fulfillment drops between the hobbyist and professional groups were not statistically significant. While a positive correlation was found between a product-focused mindset and satisfaction among professionals using AI, the overall satisfaction for AI users across the board remained significantly lower than for those who did not use AI. Furthermore, while the data collection did not directly investigate forms of creative expression outside of what is traditionally considered an art, other potential avenues of creative expression, such as in "pure" STEM areas, show evidence of having similar psychological benefits, through a similar process to traditional arts. This relationship will have to be explored more directly in future studies. These findings suggest that the integration of AI into creative tasks carries measurable psychological costs by separating the creator from the intrinsically beneficial process of making. These findings underscore the value of human-made art and suggest that the ongoing conversation surrounding AI adoption must extend beyond efficiency to account for the human experience of creation, and the value of maintaining more psychologically healthy lives.

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The Purpose of Human Art in an AI-Dominated, Product Fixated World

Introduction

Modern society often acts antithetical to the human condition. As people, we long to create and to express ourselves, all the while learning and improving ourselves within the process of creation. However, the work climate of the world promotes product over process, with the philosophy that the ends justify the means. As most of a worker's energy is directed towards getting work done, many lose the benefits of a more process focused lifestyle. With the rapid growth of generative AI, a huge and rapid shift towards only product-based valuation is signified, with people relying on passive intake rather than active creation to occupy their time. With this, the potential for an AI-fueled mental health crisis is foreshadowed. Given that much of the world is product oriented, especially with the growing phenomenon of AI providing products instantly, people who spend more time doing what they enjoy within the creative process, rather than getting the product instantly with AI use, will promote their well-being.

Literature Review

Background

The terms "creative passion" or "art" are repeatedly used throughout this paper. A creative passion is any activity whose creative process is something that one derives enjoyment from, used as an avenue of self-expression, such as expressing personal interests, expressing emotions, or expressing curiosity. This paper examines art through this lens. This may range from arts that are more traditionally considered as an art, such as painting, but also includes areas within STEM (science, technology, engineering, and math) or humanities, particularly with "pure" areas, such as pure math or pure science, which focus on more theoretical and fundamental discoveries, rather than "applied" areas, which focus on more tangible problems.

This connection between “traditional arts” and STEM will be discussed more in a later section of the literature review, and while this paper mainly focuses on traditional arts, this connection allows for the extrapolation of concluded principles, along with guiding further research to more concretely confirm these principles. This paper also makes use of the term well-being, which is characterized by emotional, cognitive, and social health, shown through aspects of mental stimulation and activation, less depressive symptoms, lower fatigue, close associates that one feels comfortable speaking to, and generally higher levels of self-described happiness and satisfaction.

With the rise of generative AI comes many usages. From these uses, many people have adopted the use of AI as a tool to create products instantly, generating images and products at a pace humans cannot compare to (Chui et al., 2023). The adoption of generative AI has grown rapidly across industries, with the potential to add \$2.6–\$4.4 trillion to the global economy, fundamentally altering work (Chui et al., 2023). With its acceleration of image and music creation, it effectively separates the product of human creativity from the labor usually required to produce it (Lee, 2022). AI reduced the labor requirements for many creative tasks, lowering the barrier of entry, but also challenges the economic viability of professional creative work (Zylinska, 2024). Due to the increasing accessibility of AI, even hobbyist creators may increasingly substitute personal creative work with generative AI (Lee, 2022; Zylinska, 2024).

For decades, creativity has been evaluated through an economic lens, but this framework has been increasingly challenged (Lee, 2022). People tend to appreciate human work more, not based on the end product, but based purely on whether or not they believe it to be created by a human or not (Lee, 2022). The rise of AI generated creative products necessitates a

reconsideration of why professional creatives should be funded, and what value human work holds beyond the end product (Lee, 2022).

Art therapy and the psychological benefits of creative engagement

Engagement in creative activities, including visual arts, writing, and music, has been consistently shown to improve emotional well-being, reduce stress, increase self-confidence, and enhance coping skills (Jean-Berluche, 2024). Nonwork creative activities specifically have been linked to better evening recovery, reduced fatigue, and improved emotional well-being, with self-expression being the key mediating variable (Berkemeyer et al., 2024). Art therapy, making art in therapeutic contexts, has demonstrated the ability to alleviate symptoms of mental illness, foster social connection, and convey subconscious feelings (Shukla et al., 2022). Artists surveyed in Australia, despite earning below average incomes, reported life satisfaction comparable to the general population, suggesting the idea that inherent rewards of creative work can compensate for economic hardship (Throsby & Petetskaya, 2017).

Visual arts, beside emotional control and psychological resilience, also stimulates cognitive processes such as flexible thinking, working memory, and longer attention spans (Barnett & Vasiu, 2024). Brain imaging studies have shown that both passive and active engagement with art activates regions of the brain associated with emotional regulation, such as the mPFC and amygdala (Barnett & Vasiu, 2024). Research on social media, when distinguishing between active users and passive use, shows that active use, creating and sharing, is more conducive to positive mental health outcomes, while passive use, consuming, is more correlated with negative mental health outcomes (Verduyn et al., 2021). Using art for emotional expression rather than to distract allowed for more lasting psychological recovery and improvement, suggesting the process of creating to be integral to its therapeutic value (Barnett & Vasiu, 2024).

Active creation has a significant indirect effect on well-being through the "Experience of Creation" (Lange et al., 2018). The distinction between active and passive use and art also applies more generally beyond social media, where the most well-being benefits are derived from active art, creating and self-expression (Verduyn et al., 2021; Berkemeyer et al., 2024).

Connections between traditional considerations of “art” and STEM disciplines

Cognitively stimulating activities, including education and intellectually challenging jobs, are correlated to higher cognitive functioning throughout life, benefits persisting through adulthood (Mather, 2020). A lifestyle that is more intellectually stimulating, such as with engagement in complex, cognitively stimulating creative tasks, is associated with better cognition (Mather, 2020).

Art therapy mainly focuses on traditional arts as an avenue of creative expression; however, through examining other mediums of creative expression, such as disciplines in STEM, these benefits may be shown more broadly. The intrinsic motivation behind creative work, with passion and persistence, was identified as the one of the most important factors in sustaining an artist’s career, showing the potential higher benefits of passion driven work (Throsby & Petetskaya, 2017).

In the process of creation, experiences during creation were categorized into themes such as "encountering oneself," gaining a better understanding of the self (Lange et al., 2018). Both artists and scientists rely heavily on curiosity and creativity (Parkin et al., 2016). Art provides a physical, tactile, and intuitive way to interpret the processes of the universe, and can enhance spatial intelligence aiding in understanding abstract scientific concepts (Parkin et al., 2016; Segarra et al., 2018). STEAM (science, technology, engineering, arts, and mathematics)

initiatives help make science more accessible, relatable, and inclusive for students, often by casting them in the roles of both artist and scientist (Segarra et al., 2018).

"Joyous exploration," a facet of curiosity focused on the pleasurable experience of the exploration process itself, is strongly linked to positive psychological well-being and lower depression (Priemysheva, 2024). Curiosity interacts significantly with goal attainment to increase subjective well-being over time (Priemysheva, 2024). The sustained effort and process of striving toward a goal enhances life satisfaction, showing that the psychological value often lies in the effort rather than just obtaining the final product (Priemysheva, 2024).

Psychological detriments of AI use instead of creative engagement

Daily active exploration predicts growth-oriented behaviors, greater meaning in life, and long-term life satisfaction (Priemysheva, 2024). In contrast, purely hedonistic or passive behaviors do not predict day-to-day changes in meaning or life satisfaction, which mirrors the psychological detriments of passive AI consumption versus active art creation (Priemysheva, 2024). Similar to how passive social media consumption leads to harming well-being, by replacing personal creation with looking at an AI generated output may also reduce or potentially erase the psychological benefits that active creation provides (Verduyn et al., 2021).

If the value of creativity is understood through the human experience of creating, then generative AI use devalues it, regardless of whether or not the end product appears to have the same quality, which is also reinforced by how people prefer some art over others, based purely off of if they think it was made by a human (Lee, 2022). The historical understanding of the creative industry has placed too much focus on the economic product, and not enough effort into understanding the noneconomic benefits of it (Lee, 2022; Throsby & Petetskaya, 2017).

Beyond the loss of creative benefits, overuse of AI has been linked to psychological dependency, along with addictive behaviors, ultimately resulting in emotional dysregulation (Head, 2025). Furthermore, preliminary data suggests that prolonged AI use may contribute to cognitive impairment, further countering the benefits of creative and cognitive activities (Head, 2025). These factors contribute to idea that AI use can become dangerous, where, from the sources reviewed, it can be reasoned that usage of generative AI instead of oneself to create the products of art is inversely correlational to well-being, where the more an AI substitutes engagement with a creative passion, the lower the benefits will be (Jean-Berluce, 2024; Barnett & Vasiu, 2024; Berkemeyer et al., 2024; Head, 2025).

However, there is a gap in direct data exploring the relationship between AI use in substituting creative work, and the resulting well-being. To address this gap, a survey was conducted, examining well-being through general satisfaction, along with social, emotional, and cognitive fulfillment.

Data Collection

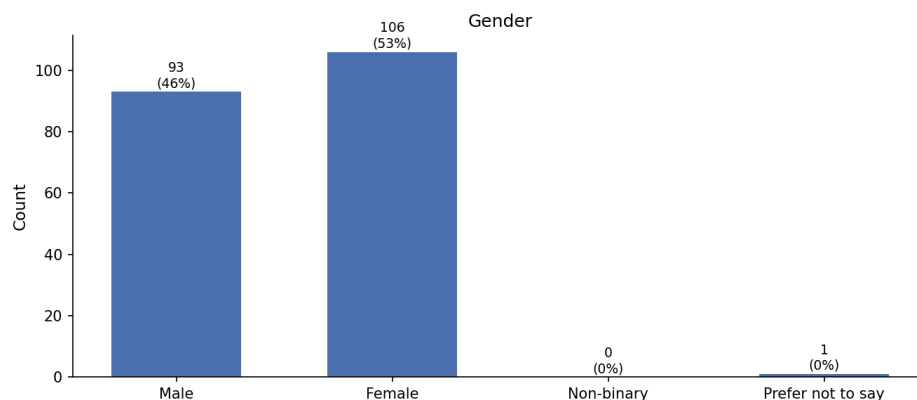
Methods

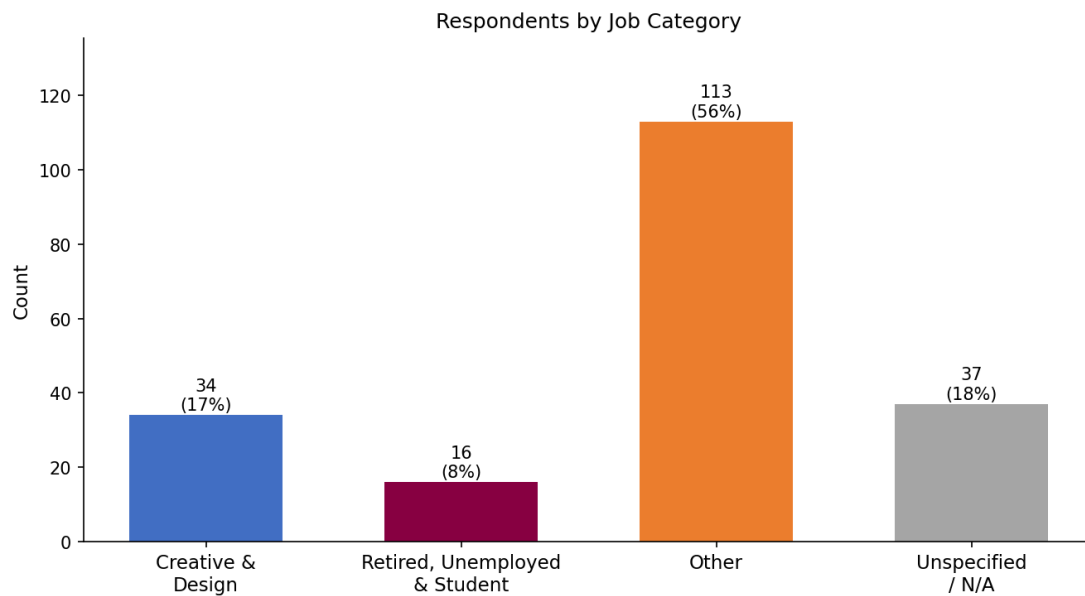
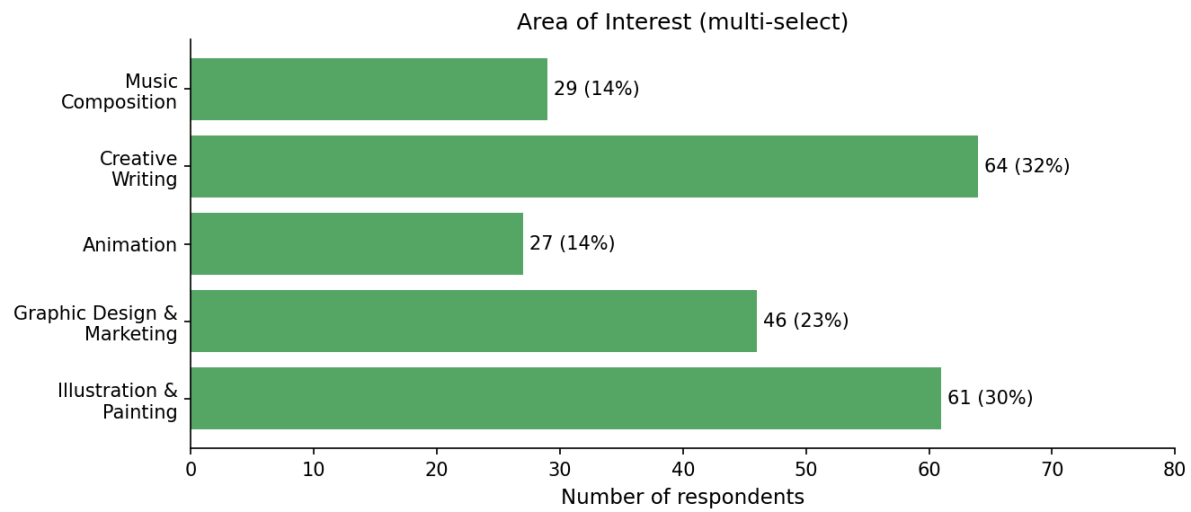
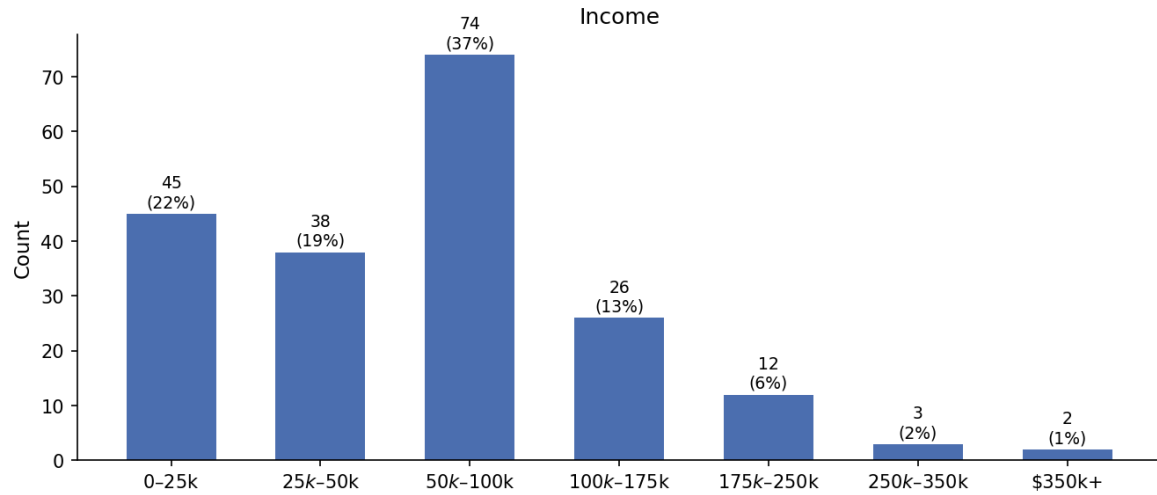
To collect data on the general impact of AI use on people, a quantitative focus with minor qualitative data would be best. Due to the various factors influencing the impact of AI use on people, the ability to have many avenues for data collected would be incredibly useful. For the same reason, the specific method of data collection used as a questionnaire. Through a questionnaire, data could be collected from a general population en masse. Furthermore, the responses to questions could be graphed and analyzed numerically, while also allowing for more qualitative questions if necessary. The target population was selected to be working adults, as the impact of having a creative outlet is heavily relevant to the idea of a work-life balance, but the

benefits of AI use also show up most clearly within working populations, able to speed up processes.

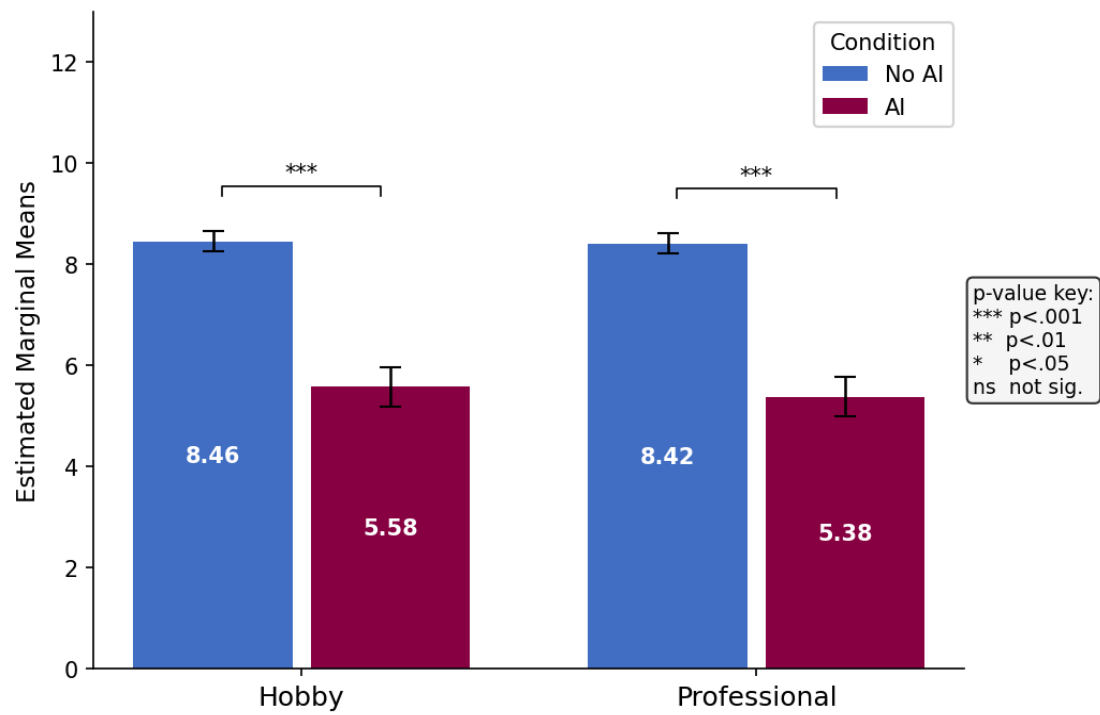
Within the questionnaire, each respondent would be separated randomly into 4 categories, a 2x2 between AI used versus no AI, and hobby versus professional. For each category, a scenario would be described, where the respondent imagines themselves as either a hobbyist or professional illustrator, and either using AI or not using AI. To get approximately 50 respondents per category, 200 total responses were collected, and after reading each scenario, they would all answer the same questions. The scenarios focus on a single discipline to minimize outside influences, being illustration due to the impact of generative AI on it. Despite the narrower scenario description, conclusions drawn should be able to be applied more broadly, across anything that can be considered to contain the creative process, directing future research to observe those other disciplines. From this, effective comparisons could be made between each scenario, of working vs. nonworking, and AI vs. no AI. From each person, their self-described emotional, social, and cognitive fulfillment, along with general satisfaction was gauged on a 1-10 scale. Furthermore, various demographic data were collected, enabling more comparisons, such as gender and income.

Data

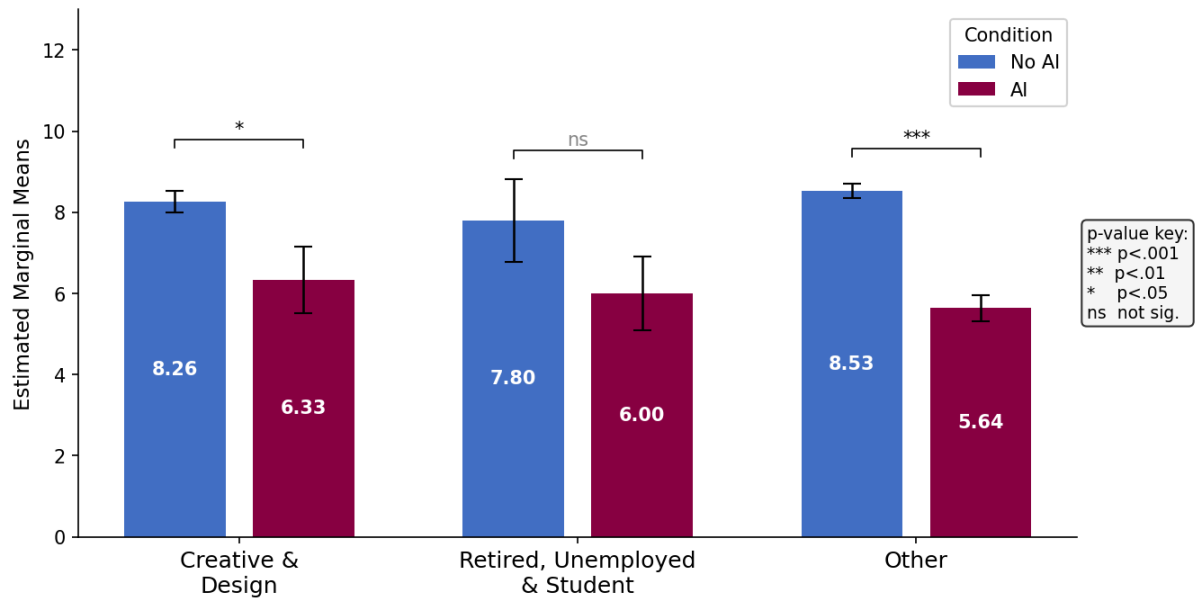


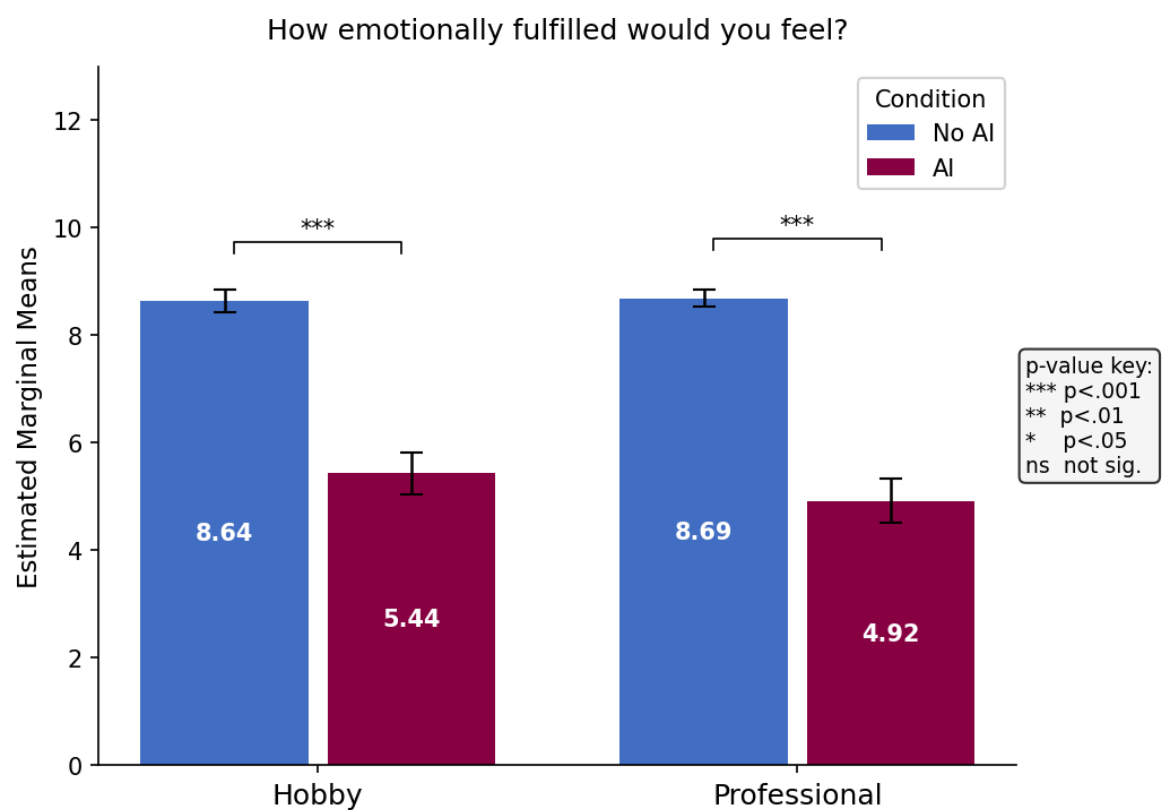
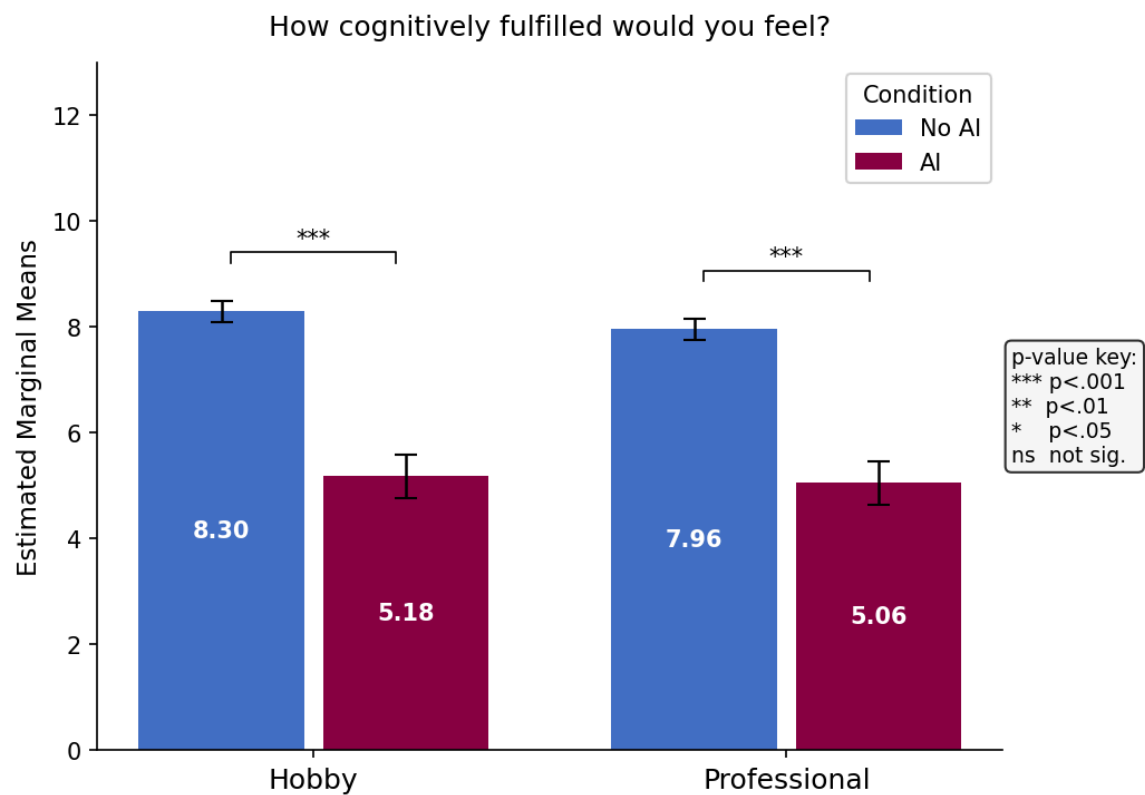


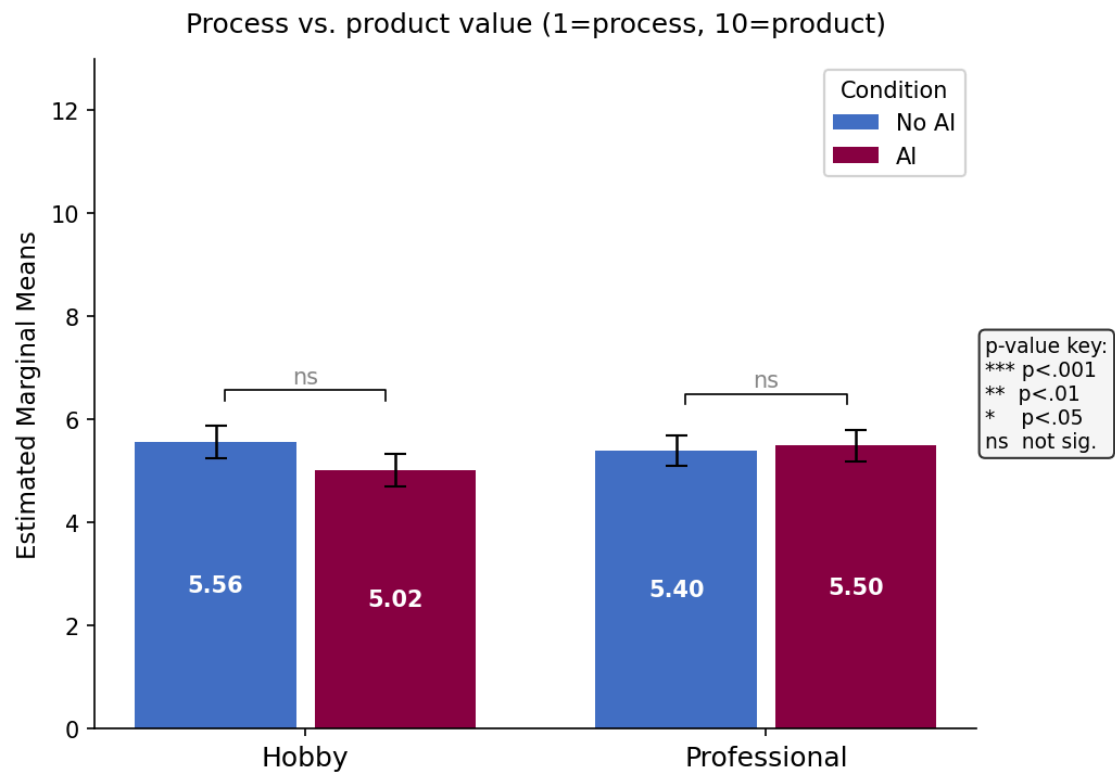
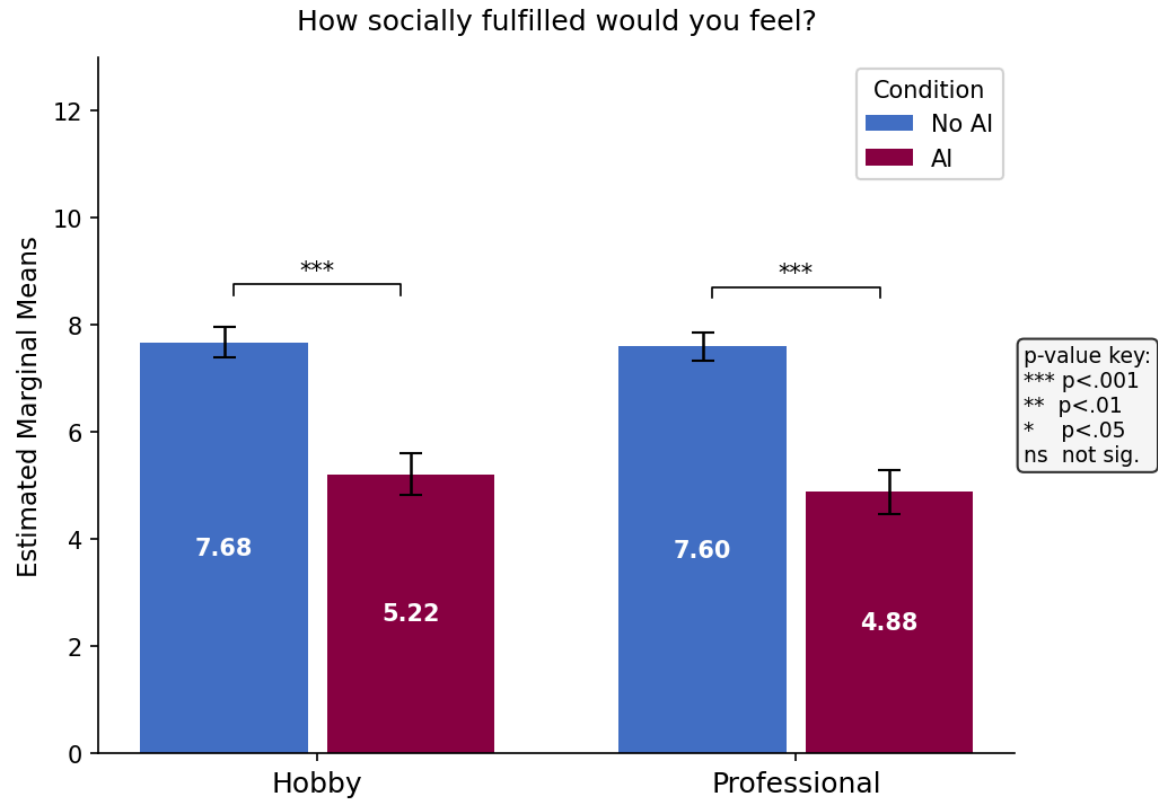
How satisfied would you feel with your work?

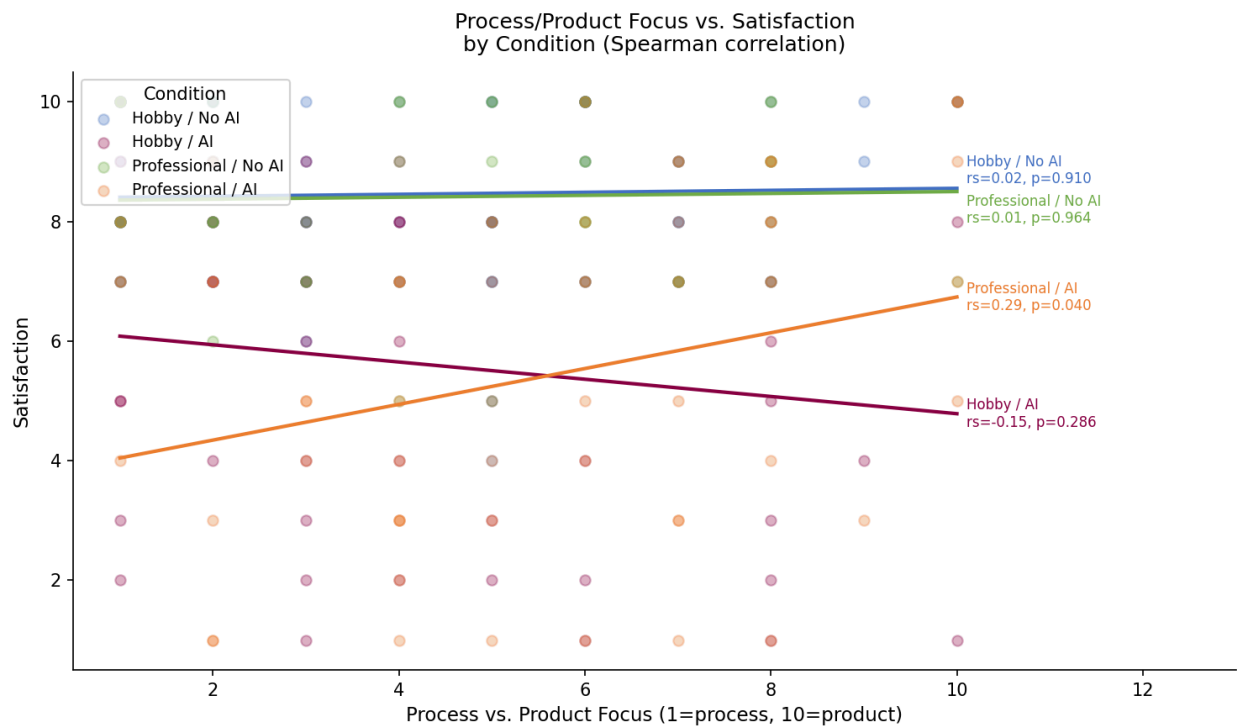
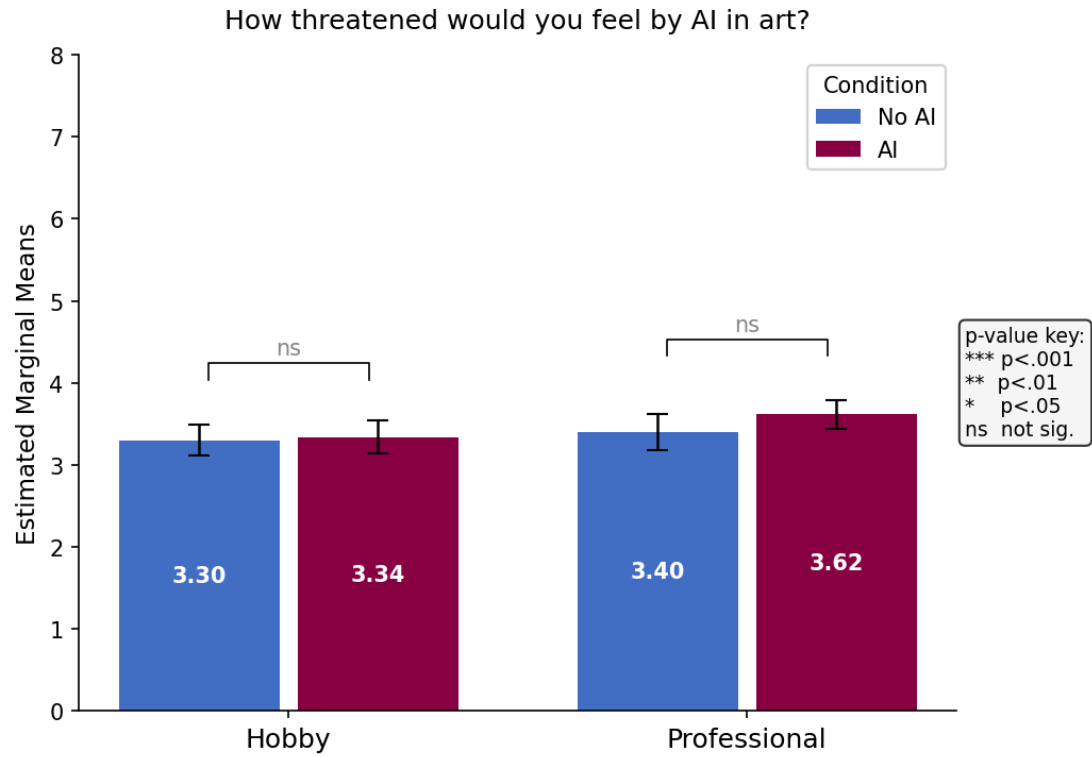


How satisfied would you feel with your work? (by Job Category)









Analysis

Trend #1: Across emotional fulfillment, social fulfillment, cognitive fulfillment, and general satisfaction, the respondents in the AI group reported significantly lower scores. For emotional (Hobby: 8.64 vs. 5.44; Professional: 8.69 vs. 4.92, $p<.001$), social (Hobby: 7.68 vs. 5.22; Professional: 7.60 vs. 4.88, $p<.001$), and cognitive fulfillment (Hobby: 8.30 vs. 5.18; Professional: 7.96 vs. 5.06, $p<.001$), in all three measures, the difference between No AI and AI conditions was statistically significant at $p<.001$, both in hobby and professional groups.

Trend #2: While slight numerical differences exist, and the mean value of fulfillment does tend to be lower in the professional group compared to the hobby group, none reached statistical significance. For example, satisfaction scores in the no AI condition were nearly identical (Hobby: 8.46, Professional: 8.42), as were AI condition scores (Hobby: 5.58, Professional: 5.38). This pattern held across emotional, social, cognitive, and threatened measures, where hobby and professional means differed by less than 0.5 points in every case and no pairwise comparison between the two groups reached $p<.05$.

Trend #3: As shown in the scatter plot along with the resulting linear regression, a positive Spearman correlation was found between product focus and satisfaction ($r_s=0.29$, $p=0.040$) among the AI group, which was also the only condition reaching statistical significance, showing that an increase in the focus on products correlates to higher satisfaction when using AI for a job. While it did not reach statistical significance, the Hobby/AI condition showed a slight negative trend ($r_s=-0.15$, $p=0.286$), with increases to product focus potentially correlated with lower satisfaction when using AI in a hobby. Both No AI groups were near-identical, showing near-zero correlations (Hobby/No AI: $r_s=0.02$, $p=0.910$; Professional/No AI: $r_s=0.01$, $p=0.964$), indicating no meaningful relationship between process/product

orientation and satisfaction when AI is not involved. However, the satisfaction in the no AI group was generally higher across the board when compared to the AI group.

Discussion

The data suggests that the integration of AI into the creative process carries several psychological costs, regardless of whether or not art is done professionally or as a hobby. There is a consistent and statistically significant drop in emotional, social, and cognitive fulfillment, along with general satisfaction, across both hobby and professional groups when AI is used. Although there may be a potential slight downwards correlation from hobby to professional groups, it is not statistically significant. This points to the idea that the beneficial effects of creative work are not significantly impacted by whether or not the work is done for profit, reflecting something more fundamental in the derivation of meaning from creative work. Furthermore, while less AI use showed beneficial results regardless of reported focus on product or process, the inherent act of AI use may put emphasis on product, and vice versa for AI non-use. If the user was able to select their level of AI use based on their personal preference, along with indicating level of focus on obtaining products, more correlations may have been drawn between product and process focused AI use versus non-use in the creative process, which may be a valuable avenue in future research.

The divergence in the scatter plot, where product-focused professional AI users showed a significant positive correlation with satisfaction while product-focused hobbyist AI users trended negatively, shows a significant impact on mindset on the benefits and disadvantages of AI use. When a professional artist uses AI, if they are process focused, they will have significantly worse satisfaction compared to if they prioritized the product. For hobbyist artists, a weak trend in the opposite direction is suggested, however, a larger data set is required for it to become statistically

significant. Ultimately, despite the positive correlation between product-focus and AI use, the satisfaction is still less than when AI is not used, across the board. Future research could examine this distinction more directly by measuring intrinsic versus extrinsic creative motivation as a variable, or by replicating the study with populations who use AI tools regularly in their actual practice rather than in a hypothetical scenario. For practitioners, educators, and policymakers navigating AI adoption in both creative industries and all industries affected by AI use, these findings offer early evidence that the conversation around AI integration should extend beyond questions of quality or efficiency, and account for the human experience of making.

Conclusion

This paper aimed to explore the impact of AI integration into the creative process on the subjective well-being and fulfillment, along with if it differs between certain types of artists. Creative engagement is known to produce significant and measurable socioemotional and cognitive benefits. Passive engagement with products, especially in social media, has shown to counteract these benefits. AI involvement is correlated with a consistent and significant reduction in satisfaction, along with socioemotional and cognitive fulfillment. The distinction between hobbyist and professional engagement with art produced few significant differences, showing that the psychological benefits of art-making arise more from the medium rather than the goal.

Although the data collected primarily reflects traditionally considered creative industries, the creative, social, and cognitive benefits of art can extend to any field that has been shown to act as a medium of cognitive activation and expression. As a result, if this study were to be replicated for more fields that use human ingenuity, such as in programming or mathematics, which is seeing a rise in AI use, it is possible that similar trends may emerge. Or, the trends may

diverge due to more emphasis on product over process, especially in software engineering. Due to this, separating pure STEM and applied STEM when discussing AI use is also necessary. A repeat of this survey across several fields with differing focuses on process vs product may show trends that illustrate more precisely in which fields AI is most beneficial to be used, and where it is least beneficial to be used.

These findings provide evidence based support for the idea that human-made art continues to be valuable, even as AI tools become more accessible and convenient. This is especially useful for those who rely on creative hobbies for emotional recovery and well-being, along with art therapists who use these techniques. These findings also contribute to the ongoing conversation on the noneconomic purpose of creative work, and what are the disadvantages of replacing human labor besides raw output.

These considerations of STEM as a method of creative engagement, or STEM itself as an “art,” as discussed in the literature review, direct future studies to also consider AI’s place in STEM when STEM can act as an art, and to what extent AI should be used to substitute the process of discovery. As was mentioned, this could look at the differences between participants in pure math versus applied math, or pure science versus applied science, or those involved in programming and software creation, and what effect AI use has on both the benefits to well-being of participants, alongside quality and impact of the results.

More research is needed to determine the impact of process vs. product orientation on well-being, along with the other benefits of creative engagement. More research is also necessary to determine more directly the effect of AI use on various other fields of creative engagement besides the traditional arts.

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Appendices

Questionnaire Questions:

<https://docs.google.com/document/d/18sZ-ARogFVjud6Rvn6wXIxxnoTZnrhWW/edit?usp=sharing&ouid=110044103503263461664&rtpof=true&sd=true>

Questionnaire Responses:

https://docs.google.com/spreadsheets/d/1tSY3bRK5XFgxDMb-Zbpb0axdk7_eiVJh/edit?usp=sharing&ouid=110044103503263461664&rtpof=true&sd=true